SQL
Exercise 2

```
12  Q01
13  SELECT distinct department
14  FROM exercise2.BFG.students;
15
```

Add parameter

Table

	department
1	IT
2	HR
3	FINANCE

```
16  Q02
17  SELECT department, ···
19  FROM exercise2.BFG.students
20  GROUP BY department;
```

Add parameter

Table

	department	avg_age
1	IT	20.5
2	HR	22
3	FINANCE	23

```sql
Q03
SELECT department,
       COUNT(*) AS student_count
FROM exercise2.BFG.students
GROUP BY department
HAVING COUNT(*)>1;
```

Add parameter

Table

	department	student_count
1	IT	2
2	HR	2

```sql
Q04
SELECT student_id,
       name,
       age,
       department
FROM exercise2.BFG.students
WHERE age BETWEEN 21 AND 23;
```

Add parameter

Table

	student_id	name	age	department
1	2	Bob	22	HR
2	3	Charlie	21	IT
3	4	Diana	23	FINANCE
4	5	Eve	22	HR

```
36  Q05
37  SELECT student_id, ···
41  FROM exercise2.BFG.students
42  WHERE department IN ('IT','HR')
43  AND age>21;
44
```

Add parameter

Table

	student_id	name	age	department
1	2	Bob	22	HR
2	5	Eve	22	HR

```
59  Q06
60  SELECT department,
61  ·····SUM(credits) AS total_credits
62  FROM exercise2.BFG.courses
63  GROUP BY department
64  HAVING SUM(credits) > 5;
65
```

Add parameter

Table

	department	total_credits
1	IT	11

```sql
Q07
SELECT course_id,
       course_name,
       department,
       credits
FROM exercise2.BFG.courses
WHERE credits<>4;
```

Add parameter

Table

	course_id	course_name	department	credits
1	101	SQL Basics	IT	3
2	104	Excel	FINANCE	2
3	105	Statistics	HR	3

```sql
Q08
SELECT course_id,
       course_name,
       department,
       credits
FROM exercise2.BFG.courses
ORDER BY credits DESC
LIMIT 3;
```

Add parameter

Table

	course_id	course_name	department	credits
1	102	Python	IT	4
2	103	Data Science	IT	4
3	101	SQL Basics	IT	3

```sql
Q09
SELECT MAX(grade) AS max_grade,
       MIN(grade) AS min_grade,
       AVG(grade) AS avg_grade
FROM exercise2.BFG.enrollments;

```

Add parameter

Table

	max_grade	min_grade	avg_grade
1	90	78	84.6

```sql

Q10
SELECT course_id,
       COUNT(*) AS enrollment_count
FROM exercise2.BFG.enrollments
GROUP BY course_id;
```

Add parameter

Table

	course_id	enrollment_count
1	101	1
2	102	1
3	103	1
4	104	1
5	105	1

```sql
Q11
SELECT department,
       SUM(salary) AS total_salary,
       SUM(bonus) AS total_bonus
FROM exercise2.BFG.salaries
GROUP BY department;
```

Add parameter

Table

	department	total_salary	total_bonus
1	IT	122000	10500
2	HR	60400	7500
3	FINANCE	70000	6000

```sql
Q12
SELECT department,
       AVG(salary) AS avg_salary
FROM exercise2.BFG.salaries
GROUP BY department
HAVING AVG(salary) > 55000;
```

Add parameter

Table

	department	avg_salary
1	IT	61000
2	FINANCE	70000

```sql
Q13
SELECT employee_id,
       name,
       salary,
       bonus,
       (salary + bonus) AS total_compensation
FROM exercise2.BFG.salaries
WHERE (salary + bonus) > 60000;
```

Add parameter

Table

	employee_id	name	salary	bonus	total_compensation
1	1	Tom	60000	5000	65000
2	3	Spike	70000	6000	76000
3	4	Tyke	62000	5500	67500

```sql
Q14
SELECT department,
SUM(budget) AS total_budget,
AVG(budget) AS avg_budget
FROM exercise2.BFG.projects
GROUP BY department
HAVING AVG(budget) > 70000;
```

Add parameter

Table

	department	total_budget	avg_budget
1	IT	270000	135000
2	Finance	80000	80000

```sql
Q15
SELECT project_id,
project_name,
department,
budget
FROM exercise2.BFG.projects
WHERE budget BETWEEN 50000 AND 120000
  AND department <> 'Marketing';
```

Add parameter

Table

	project_id	project_name	department	budget
1	1	AI App	IT	120000
2	2	Payroll System	Finance	80000
3	5	HR Portal	HR	50000